EXPRIMENT-9

9.) To decrypt the given Playfair cipher using a C program.

AIM:

Aim: To write a Python program to decrypt the given PT-109 message using the Playfair cipher with the key ROYAL NEW ZEALAND NAVY.

PROGRAM:



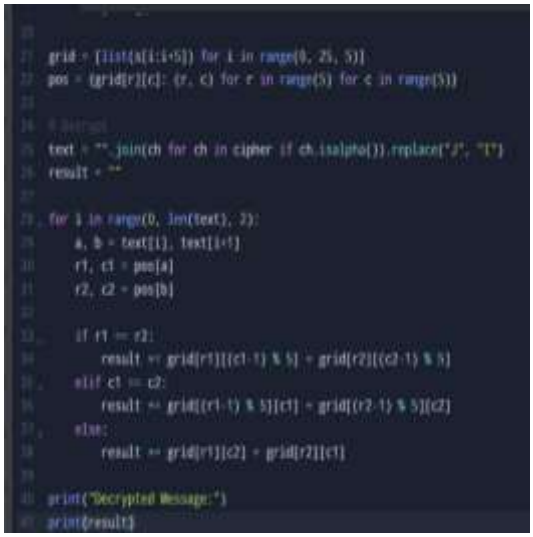
```
key = "ROYAL NEW ZEALAND NAVY"
cipher = """KUIFY UREBE ZWEHE WRYTU HEYFS
KREHE GOYFI WTTTU OLKSY CAJPO
BOTEI ZONTX BYBNT GONEY CUZAR
GDSSN SXBOU YWRHE BAAHY USEDQ"""

import string

s = ""
for ch in key.upper():
    if ch.isalpha():
        ch = ch.replace("J", "I")
        if ch not in s:
            s += ch

for ch in string.ascii_uppercase:
    if ch != "J" and ch not in s:
        s += ch

grid = [list(s[i:i+5]) for i in range(0, 25, 5)]
pos = {grid[r][c]: (r, c) for r in range(5) for c in range(
```



```
grid = [list(s[i:i+5]) for i in range(0, 25, 5)]
pos = {grid[r][c]: (r, c) for r in range(5) for c in range(5)}

text = "".join(ch for ch in cipher if ch.isalpha()).replace("J", "I")
result = ""

for i in range(0, len(text), 2):
    a, b = text[i], text[i+1]
    r1, c1 = pos[a]
    r2, c2 = pos[b]

    if r1 == r2:
        result += grid[r1][(c1-1) % 5] + grid[r2][(c2-1) % 5]
    elif c1 == c2:
        result += grid[(r1-1) % 5][c1] + grid[(r2-1) % 5][c2]
    else:
        result += grid[r1][c2] + grid[r2][c1]

print("Decrypted Message:")
print(result)
```

OUTPUT:

```
Decrypted Message:
PTBOATONEOWENINELOSTINACTIONINBLACKESSSTRAITTWOMILESSWMERESUCOVEXCREWOFTWELVEXREQUEST
ANYINFORMATIONX
```

RESULT:

Thus, frequency analysis was used to decrypt the simple substitution cipher.